Development of a System for Detecting AI-Generated Images Using Deep Learning Methods

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1 Introduction

Generators change, and images are often compressed or resized before people share them. A detector therefore needs to handle pictures beyond its training examples. Our project asks how well models trained on several generators detect images from another one. We compare two pretrained models in a preliminary experiment before carrying out the larger study with matched data.

2 Literature Review

2.1 Why the test data matters

Foo et al. [1] and Chen et al. [2] review the development of generative models, including diffusion-based image generation. Detection surveys group approaches into CNNs, transformers and forensic features [3, 4]. Generalization and processing robustness recur as unresolved problems [7, 8]. Broader deepfake reviews describe substantial differences in datasets and evaluation conditions [9–12]. For our project, this makes a shared independent test more useful than a ranking of published accuracy figures.

Coccomini et al. [5] compare CNNs and CLIP-based features on Stable Diffusion and GLIDE images. Transfer is asymmetric, so the direction from one source to another needs its own test. Their results also vary with image content, which informs our plan to balance categories. Bayram [6] finds a gap between CIFAKE and external evaluation. Rout et al. [18] report an internal-to-external drop in face detection, although their face-specific setting leaves performance on the varied objects in GenImage unresolved. We therefore reserve one generator and measure false positives on real images separately.

2.2 Choosing a comparison we can complete

Lamichhane [13] studies a vision transformer, while Saskoro et al. [14] combine CNN experts trained for different generators. The expert system has several components to implement. We chose a smaller initial comparison of a conventional CNN and frozen pretrained features on that basis. Sanders et al. [21] compare a compact CNN with ResNet-18 and test new object categories within SDXL images. They leave post-processing outside the experiment. Their category-transfer test consequently answers a different question from our test on a new generator.

Man and Cho [15] combine semantic and frequency information; Yu and Xu [19] add texture descriptors. Feature fusion could be a later extension, with an ablation to measure its contribution under our split. The published comparisons do not isolate that contribution for this dataset. Studies of AI-generated artwork examine transformers, explanations and spectral features [17, 20], but their results are specific to art. Yumlembam et al. [16] reject predictions using uncertainty, removing some correct answers along with errors. Evaluating such rejection requires both error filtering and retained coverage. We first need to establish the baseline errors before adding these components.

The gap we address is transfer to a new generator together with robustness to processing under a shared, audited protocol. Ojha et al. [25] show that frozen vision-language features can transfer to unfamiliar generators, which motivates our CLIP baseline. Its scores will be measured on our own data. Grommelt et al. [27] show that compression and native-size cues can affect apparent detection performance. Those cues need attention before a model difference can be attributed to generation artifacts.

3 Methods and Materials

3.1 Questions and comparison

RQ1 asks how well the two detectors identify images from an unseen generator while limiting false positives on real images. RQ2 asks how JPEG compression and resizing change F1 and false-positive rate. We compare a fine-tuned ResNet-50 with a frozen CLIP image encoder and a trained linear classifier on the same image IDs. Mahara and Rishe [4] inform the choice of spatial and vision-language representations. Xu et al. [7] distinguish transfer from processing robustness, while the external tests in Coccomini et al. [5] and Bayram [6] motivate reserving an unfamiliar source.

3.2 Data and leakage controls

The full study will use GenImage [26]. SD 1.4, GLIDE and BigGAN will supply development images, with Midjourney held out. The targets are 12,000 training, 2,000 validation, 2,000 seen-test and 4,000 unseen-test images, each half real. The release description reports approximately 162 training and six test generated images per class for these sources across 1,000 classes. Our unseen target thus needs 2,000 generated images from a nominal 6,000-image Midjourney test pool. We still need to audit the downloaded archives and establish the counts after readability checks, deduplication and matching. Training and validation will come from official training folders; reserved tests will use official validation folders.

The manifest will record source, label, category, original partition, dimensions, actual codec, JPEG tables, chroma subsampling and split. We will use exact hashes and perceptual checks to screen duplicates across splits, then balance real and generated records within available category and size bins. Following Grommelt et al. [27], we will select real JPEGs with common compression settings and encode generated images with those settings before the shared RGB preparation. Real files will not receive that extra pass. A development-only before/after encoding check will measure class-specific score changes. This checks sensitivity to the encoding step; earlier compression history can still supply shortcuts after matching. Final inputs will be 224×224 pixels, stored losslessly after preparation.

3.3 Training and the preliminary run

ResNet-50 will use torchvision ImageNet1K V1 weights and a binary output, following He et al. [22] and Wang et al. [24]. CLIP ViT-B/32 will use published OpenAI weights with its image encoder frozen [23, 25]. Both classifiers use binary cross-entropy and Adam. We select the checkpoint with the best validation F1. Test preparation is deterministic, with model-specific normalization after the shared image preparation. Development work will establish the full-study learning rates, batch size and stopping rules.

The full comparison will use sampling seed 42 and training seeds 42, 43 and 44, retaining individual runs as well as their means and sample standard deviations. Colab is the planned platform.

Pilot configuration. The feasibility experiment uses a fixed 480-image Tiny-GenImage mirror sample [28], with BigGAN and GLIDE for development and Midjourney held out. Each model is trained with seeds 42, 43 and 44 on identical splits. ResNet uses one epoch, batch size eight and learning rate 10^{-4}; the frozen CLIP encoder's linear head uses 40 epochs, batch size 32 and learning rate 10^{-3}. The runs use six PyTorch threads on an AMD EPYC 9V74 CPU, with no GPU. Native-size, JPEG-table and category matching are outside this pilot.

3.4 Metrics and paired testing

Generated images are the positive class. A false positive is a real image labelled generated. At the primary threshold of 0.5 we calculate

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{Precision} = \frac{TP}{TP + FP}, \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2TP}{2TP + FP + FN}, \tag{2}$$

$$\text{FPR} = \frac{FP}{FP + TN}. \tag{3}$$

Undefined precision or F1 is assigned zero. We also calculate AUROC and average precision from continuous scores. Full-study success requires mean original unseen F1 ≥ 0.80, mean FPR ≤ 0.10 on both original tests, and a seen-to-unseen F1 drop ≤ 0.10. We will report how many seeds meet all the goals.

A paired, label-by-source stratified bootstrap with 1,000 resamples gives percentile 95% intervals. Each resample uses the same image IDs for both models and all seeds. Metrics are calculated per seed before averaging. The intervals describe image sampling conditional on the trained models; variation between training runs is reported separately. A meaningful model gap requires at least 0.03 F1, an interval excluding zero and an FPR within the target for the model with higher F1.

For the primary original-unseen F1 comparison, we also use a two-sided paired permutation test. Within each image, model outputs are exchanged together across seeds. We recalculate the mean F1 difference for 9,999 permutations. The Monte Carlo p-value is $(1 + k)/(1 + 9,999)$, where k counts absolute permuted differences at least as large as observed. The statistical significance level is $p < 0.05$. Other condition-specific tests are descriptive. Images are the paired units in this test; the three seeds are repeated training runs.

3.5 Robustness and threshold sensitivity

We will test fixed checkpoints on original images, JPEG quality 65 and independent downsampling to 112×112 pixels [24, 26]. For RQ2, $\Delta F_1 = F_1(\text{processed}) - F_1(\text{original})$ and $\Delta\text{FPR} = \text{FPR}(\text{processed}) - \text{FPR}(\text{original})$. Less negative F1 change and a smaller FPR increase indicate greater robustness. We consider both measures when they disagree. Paired intervals use the common evaluable IDs.

The prespecified sensitivity analysis compares threshold 0.5 with a validation-selected threshold. Each model is evaluated on validation F1 at thresholds 0.05 to 0.95 in steps of 0.05; ties select the larger threshold. We freeze the choice before evaluating unseen labels. The primary comparison retains threshold 0.5. This analysis changes the decision threshold without retraining or selecting it from test data.

3.6 Reproducibility and scope

Prediction exports contain stable IDs, labels, sources, conditions, models, seeds and scores. The archive includes training and evaluation code, the pilot manifest and run metadata. Prediction timing measures model responsiveness at batch size one after warm-up, excluding loading and image decoding. Timings are descriptive observations on a shared CPU host. The pilot records its software, hardware and elapsed times by run. Native-size matching, three-seed training and rotated generator holdouts remain full-study work. One held-out generator and uncertain pretraining exposure limit the transfer claims. Video, partial edits and authenticity certification are outside scope.

4 Results

The pilot used 480 images and training seeds 42, 43 and 44. Both detectors had lower F1 on Midjourney than on familiar generators. The original unseen CLIP-minus-ResNet difference reached the prespecified statistical significance level, while CLIP exceeded the false-positive target. Neither model met all project success criteria. RQ1 concerns unseen-generator detection and false positives on real images. RQ2 concerns the changes after JPEG compression and resizing. The matched 20,000-image study remains planned.

4.1 Data actually used and departures from the full study

We accessed the Tiny-GenImage mirror [28] on 17 October 2026 and recorded revision `89c4fe9efd0e`. The sampling frame contained the first 1,200 training and 600 validation metadata rows. Sampling seed 42 selected 480 images. The remaining 1,320 rows were outside the sample and were not cleaning removals. All selected assets downloaded and decoded successfully. SHA256 checks found no exact duplicate group. The same manifest and image hashes were used for each training seed; none of the additional runs selected new images.

Training contained 120 real images, 60 BigGAN images and 60 GLIDE images. Validation contained 40 real images and 20 from each development generator. The seen test used the same source counts as validation with different IDs. The unseen test contained 40 real images and 40 Midjourney images. Midjourney was absent from training and validation. Table 1 records these pilot counts beside the full-study targets. The reserved tests contain 160 distinct images, each evaluated in three conditions by two models across three runs.

Table 1 Preliminary sample actually used. Full-study targets are separate. The same 480 readable, exactly deduplicated images are reused for all three training seeds. Source: Tiny-GenImage mirror, accessed 17 October 2026.

Partition	Full target	Pilot n	Real	Generated
Training	12,000	240	120	120
Validation	2,000	80	40	40
Seen test	2,000	80	40	40
Unseen test	4,000	80	40	40
Total	20,000	480	240	240

The pilot used the public mirror, omitted SD 1.4 from development and retained the small sample before matching. Perceptual duplicate screening, category matching and matching of native size and JPEG tables were not completed. These departures from the full design are recorded with the manifest. The training-seed repeats changed model initialization and training order while preserving image IDs, source labels and

split assignments. Repeated scores for an image therefore remain observations on that same image.

All retrieved assets decoded as JPEG. Real-image widths ranged from 139 to 2,048 pixels and heights from 127 to 1,536. BigGAN images were 128×128, GLIDE images 256×256 and Midjourney images 1024×1024. Original compression histories and category balance remain unresolved. These measurements describe the mirror assets. The official archives still require inspection before confirming the 20,000-image target and the counts retained after cleaning and matching. No official-archive audit count is reported as an observed pilot total.

4.2 RQ1 original-image detection and transfer

Table 2 reports mean run-level metrics with sample standard deviations at threshold 0.5. On familiar generators, ResNet-50 achieved mean F1 0.972 (SD 0.018), compared with 0.870 (SD 0.017) for CLIP linear. Mean FPR was 0.058 (SD 0.038) for ResNet and 0.267 (SD 0.029) for CLIP. ResNet detected all 40 generated images in every run. CLIP detected 40, 39 and 38 for seeds 42, 43 and 44, respectively.

Table 2 Original-image metrics at threshold 0.5. Upper cells show mean with sample SD in parentheses over training seeds 42, 43 and 44; lower rows show unseen confusion counts for each run. Each test has 40 real and 40 generated images, reused across seeds. P and R are precision and recall; AUC is AUROC.

Model	Test	Acc.	P	R	F1	FPR	AUC	AP
ResNet-50	Seen	0.971 (0.019)	0.946 (0.034)	1.000 (0.000)	0.972 (0.018)	0.058 (0.038)	0.999 (0.002)	0.999 (0.002)
CLIP linear	Seen	0.854 (0.019)	0.785 (0.018)	0.975 (0.025)	0.870 (0.017)	0.267 (0.029)	0.946 (0.005)	0.946 (0.005)
ResNet-50	Unseen	0.504 (0.014)	0.556 (0.509)	0.025 (0.025)	0.047 (0.047)	0.017 (0.014)	0.487 (0.088)	0.527 (0.061)
CLIP linear	Unseen	0.458 (0.014)	0.392 (0.046)	0.158 (0.038)	0.225 (0.046)	0.242 (0.014)	0.518 (0.010)	0.519 (0.011)

Unseen counts: model	Seed	TP	FP	TN	FN
ResNet-50	42	2	1	39	38
ResNet-50	43	0	1	39	40
ResNet-50	44	1	0	40	39
CLIP linear	42	8	10	30	32
CLIP linear	43	6	10	30	34
CLIP linear	44	5	9	31	35

On the original unseen set, mean ResNet F1 was 0.047 (SD 0.047), with mean recall 0.025 and mean FPR 0.017. Its true-positive counts were two, zero and one. CLIP's mean F1 was 0.225 (SD 0.046), recall was 0.158 and FPR was 0.242. Its true-positive counts were eight, six and five. The lower part of Table 2 gives TP, FP, TN

and FN separately for each unseen run. Every count row covers the same 40 real and 40 generated images.

The original unseen mean AUROC was 0.487 (SD 0.088) for ResNet and 0.518 (SD 0.010) for CLIP. Their mean AP values were 0.527 and 0.519. Mean accuracy was 0.504 for ResNet and 0.458 for CLIP. On familiar images, mean AUROC and AP rounded to 0.999 for ResNet and 0.946 for CLIP. Table 2 also includes precision and recall, averaged after calculation for each run.

The mean seen-to-unseen F1 drops were 0.925 for ResNet and 0.645 for CLIP, exceeding the 0.10 limit. Both mean unseen F1 values were below 0.80. ResNet's FPR stayed within 0.10 on both original tests in all three runs. CLIP exceeded that limit on both tests in each run. Neither detector satisfied all project goals in any run: the count was zero of three for each baseline. These checks use the fixed primary threshold and original-image conditions.

Seen false-positive counts for ResNet were four, one and two in seed order; CLIP's were ten, twelve and ten. Individual seen F1 ranged from 0.952 to 0.988 for ResNet and 0.857 to 0.889 for CLIP. Original unseen F1 ranged from 0.000 to 0.093 and from 0.185 to 0.276, respectively. ResNet seed 43 produced no correct generated-class detection on the original unseen set. Its unseen false-positive count was one.

4.3 Paired uncertainty and statistical testing

The primary difference in mean original unseen F1, CLIP minus ResNet, was 0.178. Its paired bootstrap 95% interval was [0.052, 0.316], with permutation p-value 0.0261 (Table 3). The interval excluded zero and the p-value was below 0.05. CLIP's mean unseen FPR of 0.242 exceeded the practical target of 0.10. The observed F1 gap therefore passed the statistical test and the minimum-gap requirement, but failed the FPR requirement for a practically meaningful advantage.

Table 3 Paired differences in mean F1, CLIP minus ResNet, across three fixed training runs. Intervals use 1,000 paired image resamples; p-values use 9,999 paired permutations. Original unseen images are primary; other p-values are descriptive and unadjusted.

Test	Condition	ΔF_1	95% CI	p
Seen	Original	-0.102	[-0.159, -0.047]	0.0004
Seen	JPEG 65	-0.121	[-0.183, -0.066]	0.0003
Seen	Resize 112	-0.027	[-0.046, -0.011]	0.0016
Unseen	Original	0.178	[0.052, 0.316]	0.0261
Unseen	JPEG 65	0.171	[0.043, 0.308]	0.0405
Unseen	Resize 112	0.012	[-0.018, 0.044]	0.4139

On familiar original images, the CLIP-minus-ResNet difference was -0.102, with interval $[-0.159, -0.047]$ and p-value 0.0004. The corresponding JPEG-compressed seen difference was -0.121, with p-value 0.0003. These are secondary comparisons. The original unseen comparison remains primary; the other p-values are descriptive

and have no multiple-comparison adjustment. Table 3 retains the sign of the difference for every condition. ResNet's original unseen AUROC values were 0.409, 0.471 and 0.582 across seeds. CLIP's were 0.524, 0.523 and 0.506. ResNet precision on that set ranged from 0.000 to 1.000, with mean 0.556 and SD 0.509.

For JPEG-compressed unseen images, the mean difference was 0.171, with interval [0.043, 0.308] and p-value 0.0405. For resized unseen images, it was 0.012, with interval [−0.018, 0.044] and p-value 0.4139. The resized difference was below the 0.03 practical minimum. On familiar resized images, the difference was −0.027, also below that minimum, with p-value 0.0016. None of these processing comparisons met all the declared practical criteria.

The three completed runs permit sample standard deviations over training runs. The image-bootstrap intervals remain conditional on the three fitted checkpoint pairs, using the paired procedures in Section 3. Each test still has only 40 real images, so one changed decision changes a single run's FPR by 0.025. The three training seeds reuse those images. Individual confusion counts, across-run SDs and image-sampling intervals are reported separately, with no increase in the stated number of independent test images.

4.4 RQ2 processing changes

All reserved test images produced both processed versions successfully. No image was excluded from a paired comparison. Table 4 gives mean processed-minus-original changes and paired 95% intervals for both F1 and FPR. Figure 1 gives absolute unseen scores averaged over the three runs, with image-bootstrap intervals. The full-study reference levels remain F1 at least 0.80 and FPR at most 0.10. The processing conditions use separate transformations of the same originals.

Table 4 Processing changes: mean across training seeds, with paired image-bootstrap 95% CI below each value. Both changes are processed minus original; negative ΔF_1 and positive ΔFPR indicate deterioration. Each comparison uses the same 80 image IDs per test. Intervals condition on the three trained checkpoints.

Model	Test	JPEG ΔF_1	JPEG ΔFPR	112 ΔF_1	112 ΔFPR
ResNet-50	Seen	0.008 [-0.009, 0.022]	-0.025 [-0.050, 0.000]	-0.278 [-0.306, -0.245]	0.825 [0.733, 0.908]
CLIP linear	Seen	-0.011 [-0.046, 0.019]	0.008 [-0.067, 0.083]	-0.203 [-0.260, -0.150]	0.733 [0.591, 0.858]
ResNet-50	Unseen	0.000 [0.000, 0.000]	-0.008 [-0.025, 0.000]	0.608 [0.534, 0.667]	0.883 [0.808, 0.950]
CLIP linear	Unseen	-0.007 [-0.074, 0.069]	-0.017 [-0.075, 0.025]	0.442 [0.314, 0.568]	0.758 [0.625, 0.883]

After JPEG quality 65, ResNet's mean unseen F1 stayed at 0.047. The paired F1 change was zero in every run, giving interval [0.000, 0.000]. Its FPR change was −0.008, with interval [−0.025, 0.000]. CLIP's mean unseen F1 was 0.218, a change of −0.007

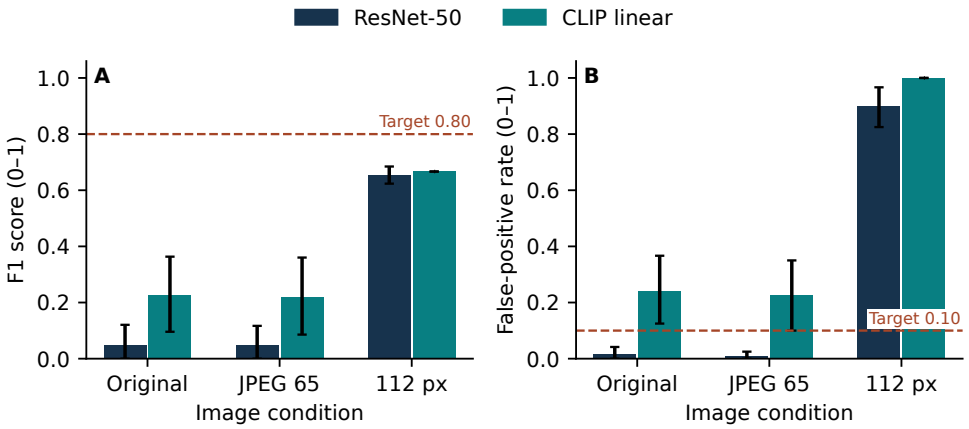


Fig. 1 Mean unseen performance over training seeds 42, 43 and 44 on 80 distinct images. Error bars are paired image-bootstrap 95% intervals conditional on the fitted models. Scores use a 0–1 scale. Dashed lines mark the full-study F1 minimum 0.80 and FPR maximum 0.10.

with interval [−0.074, 0.069]. Its FPR change was −0.017, with interval [−0.075, 0.025]. The corresponding absolute mean FPRs were 0.008 for ResNet and 0.225 for CLIP.

Downsampling increased mean unseen ResNet F1 by 0.608, with interval [0.534, 0.667], to 0.655. CLIP F1 increased by 0.442, with interval [0.314, 0.568], to 0.667. Mean FPR increased by 0.883 for ResNet and 0.758 for CLIP, reaching 0.900 and 1.000. CLIP labelled all 40 real images generated in each resized unseen run. Both models exceeded the false-positive target after this transformation.

On familiar generators, JPEG changed mean F1 by 0.008 for ResNet and −0.011 for CLIP; both intervals crossed zero. Their FPR changes were −0.025 and 0.008. Resizing reduced mean F1 by 0.278 and 0.203, with intervals [−0.306, −0.245] and [−0.260, −0.150]. Mean FPR rose by 0.825 and 0.733. All main processing intervals are shown in Table 4. The experiment covered JPEG quality 65 and downsampling to 112 pixels; no wider JPEG-quality sweep was run.

4.5 Sensitivity to the decision threshold

The validation rule selected ResNet thresholds 0.85, 0.60 and 0.30 for seeds 42, 43 and 44. It selected 0.50 for CLIP in every run. These choices used validation predictions only. Table 5 compares them with the primary threshold 0.50 on identical original test IDs. It lists each threshold and the mean test scores with sample SDs. The underlying model checkpoints and continuous scores were fixed during this comparison.

At its validation-selected thresholds, ResNet's mean seen F1 was 0.980 and mean FPR was 0.042. At 0.50 they were 0.972 and 0.058. For seed 42, the selected threshold reduced seen false positives from four to two. Seeds 43 and 44 retained their seen decisions. Each ResNet run still detected all 40 familiar generated images. The threshold choices therefore changed the reported seen averages through seed 42's real-image decisions.

Table 5 Original-image threshold sensitivity. Thresholds are listed for seeds 42, 43 and 44, in that order. F1 and FPR show mean with sample SD in parentheses. Each validation threshold is selected independently using only validation predictions. Each run evaluates the same 80 images per test.

Model	Test	Rule	t_{42}, t_{43}, t_{44}	F1	FPR
ResNet-50	Seen	Fixed	0.50, 0.50, 0.50	0.972 (0.018)	0.058 (0.038)
ResNet-50	Seen	Validation	0.85, 0.60, 0.30	0.980 (0.007)	0.042 (0.014)
ResNet-50	Unseen	Fixed	0.50, 0.50, 0.50	0.047 (0.047)	0.017 (0.014)
ResNet-50	Unseen	Validation	0.85, 0.60, 0.30	0.033 (0.028)	0.000 (0.000)
CLIP linear	Seen	Fixed	0.50, 0.50, 0.50	0.870 (0.017)	0.267 (0.029)
CLIP linear	Seen	Validation	0.50, 0.50, 0.50	0.870 (0.017)	0.267 (0.029)
CLIP linear	Unseen	Fixed	0.50, 0.50, 0.50	0.225 (0.046)	0.242 (0.014)
CLIP linear	Unseen	Validation	0.50, 0.50, 0.50	0.225 (0.046)	0.242 (0.014)

On Midjourney, ResNet's validation-selected mean F1 was 0.033 and mean FPR was zero. The true-positive counts were one, zero and one. At threshold 0.50 those counts had been two, zero and one. CLIP's seen and unseen metrics were unchanged in every run because all its selected thresholds equalled 0.50. AUROC and AP were unchanged for both models throughout the threshold comparison. There was no retraining in this analysis.

Validation contained 80 images from the development sources. The selected thresholds apply to this pilot's validation samples, with the same image IDs across runs. The future matched-data experiment will select its own thresholds under the declared rule. The fixed 0.50 results remain primary, and no threshold was chosen from unseen test performance. The sensitivity export contains individual-run scores as well as the averages in Table 5.

4.6 Observed errors

Across original unseen runs, ResNet missed 38, 40 and 39 Midjourney images. CLIP missed 32, 34 and 35. ResNet incorrectly flagged one, one and zero real images; CLIP flagged ten, ten and nine. Figure 2 shows the confusion matrices and inspected error examples for seed 42. That run made 39 unseen errors with ResNet and 42 with CLIP. The full three-run unseen counts remain in Table 2.

On seen BigGAN images (128×128 pixels), mean recall was 1.000 for both models. On GLIDE (256×256), it was 1.000 for ResNet and 0.950 (SD 0.050) for CLIP. On held-out Midjourney (1024×1024), mean recalls were 0.025 and 0.158. Each seen generator contributed 20 images per test run. Native-size and content-category effects remain unseparated in this pilot. Dimensions, source groups and individual error records are retained for later grouped checks.

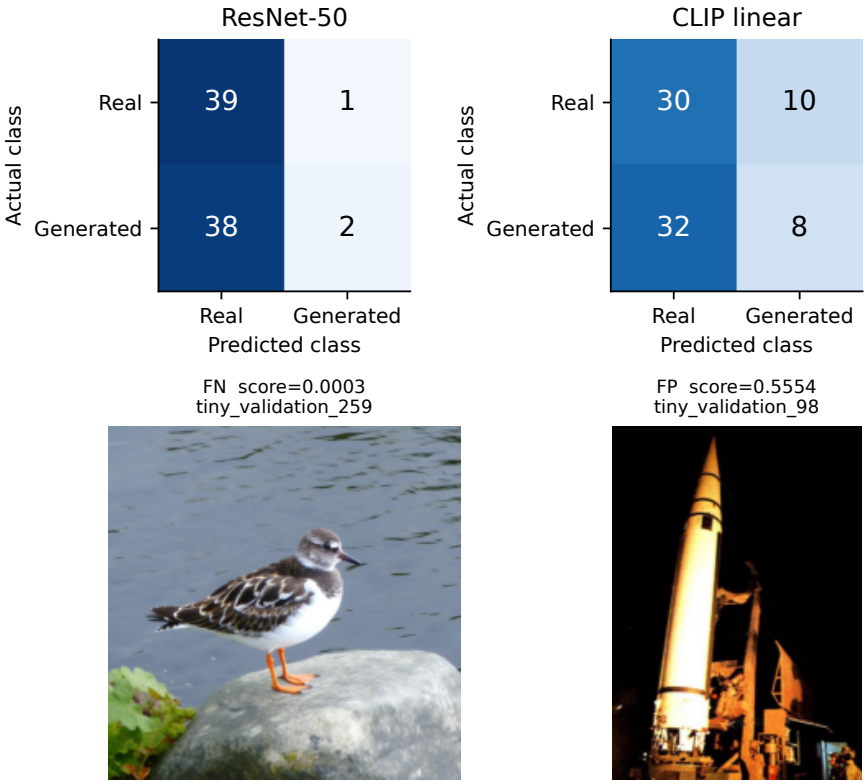


Fig. 2 Illustrative original-unseen errors for seed 42 only, threshold 0.5; all three runs are counted in Table 2. Matrix cells count images, with actual class on rows and predicted class on columns. Lower panels show the most confident ResNet false negative and CLIP false positive within those error types. Scores are generated-class sigmoid outputs. Images and labels are from Tiny-GenImage [28]; IDs identify manifest records.

The selected ResNet miss was a Midjourney-labelled bird image with generated-class score 0.000287. The selected CLIP false positive was a real-labelled image with score 0.555357. Both examples came from seed 42's original unseen predictions. The examples use dataset labels to define errors. For each indicated error type, selection used the largest score distance from 0.5. The complete export retains IDs, scores, conditions, seeds and error types for all model errors. We kept checkpoints and training settings fixed after inspection. The recorded findings include low original unseen detection, zero ResNet JPEG F1 change, unchanged CLIP threshold choices and high FPR after resizing.

4.7 Observed setup and reproducibility

All three reported runs used the AMD CPU setup in Table 6. ResNet completed one epoch per run, with training losses from 0.324 to 0.399 and validation F1 from 0.949 to 1.000. Each CLIP head completed 40 epochs. Its final-epoch training losses ranged

from 0.528 to 0.531. The selected head checkpoints were epochs 40, 21 and 34 for seeds 42, 43 and 44, with validation F1 values 0.964, 0.952 and 0.976.

Table 6 Observed three-seed setup. Preparation/training times are averaged across runs; prediction medians show the observed range. CLIP preparation includes development feature extraction. All three reported runs reuse the hash-verified image cache on the same CPU type. Times are descriptive shared-host observations; no GPU was used.

Item	Observed setting
Source and revision	`TheKernel01/Tiny-GenImage`, revision `89c4fe9efd0e`
Device	AMD EPYC 9V74; six PyTorch threads; 8 GiB container
Python and libraries	3.12.14; PyTorch 2.14.1+cpu; torchvision 0.29.1+cpu; CLIP 1.0.1
Analysis libraries	NumPy 2.5.2; Pillow 12.3.0; scikit-learn 1.8.0; SciPy 1.17.0
Seeds and optimizers	Sampling 42; training 42, 43, 44; Adam; learning rates 10^{-4} (ResNet), 10^{-3} (head)
Epochs and batches	ResNet: 1 epoch, batch 8; CLIP head: 40 epochs, batch 32
Preparation and training	ResNet 41.79 s; CLIP 11.21 s
Prediction median	ResNet 36.43 to 46.29 ms/image; CLIP 45.91 to 370.78 ms/image
Pilot elapsed by seed	42: 130.93 s; 43: 94.17 s; 44: 151.10 s
Repository	https://github.com/C0nflu3nc3/RMT (private); complete score-reproduction files in ZIP
Analysis revision	`491a65b7fbf6`; code, manifest, predictions and statistical exports

Per-run elapsed times were 130.93, 94.17 and 151.10 seconds. These runs reused the hash-verified image cache. Mean preparation/training times were 41.79 seconds for ResNet and 11.21 for CLIP, including development feature extraction for CLIP. Warmed single-image prediction medians ranged from 36.43 to 46.29 milliseconds for ResNet and 45.91 to 370.78 for CLIP. The measurements came from a shared CPU host. Software installation and subsequent statistical analysis were outside the elapsed totals.

The archive contains all 2,880 test prediction rows, validation scores, the 480-image manifest and per-run histories. The score-only reproduction command rebuilds Tables 2–5 in an isolated copy and checks them against the supplied LaTeX tables. It requires no GitHub connection, image download, model weights or PyTorch. The repository is private; the ZIP includes the required scripts and score files. The earlier Intel feasibility run is retained separately and excluded from the three-run aggregates. Official-archive auditing and the matched-data experiment remain future work.

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